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A GA-Warm-Started Mixed-Integer Nonlinear Framework for Discrete IS 800-Compliant Space-Truss Optimization

Abstract

Discrete sizing and topology optimization of steel space trusses becomes a non-smooth mixed-integer problem when catalogue sections and member buckling checks are imposed during the search. This study develops a GA-warm-started mixed-integer nonlinear optimization framework for Indian standard equal-angle trusses. A genetic algorithm first explores binary member topology and SP 6(1):1964 section indices. Elite feasible or near-feasible designs then warm-start a second integer nonlinear search in which catalogue indices are refined while IS 800:2007 stress, slenderness, and Annex D compression-capacity checks are evaluated explicitly for each candidate design. Structural response is computed using the direct stiffness method, so each candidate is assessed through analysis–design coupling rather than post-optimization section rounding. The framework is tested on 6-, 25-, and 72-bar space-truss benchmarks and compared with deterministic, mathematical-programming, and metaheuristic baselines over 20 independent stochastic runs. The proposed hybrid reproduces the 6-bar benchmark best weight of 101.17 kg, reduces the 25-bar GA best baseline by 31.8% to 248.10 kg, and reduces the 72-bar GA best baseline by 34.6% to 241.67 kg. The results show that warm-started integer nonlinear search can improve catalogue-constrained truss designs while maintaining direct code-based feasibility screening. The study also identifies cases where lighter penalty-based designs require independent post-optimization verification, emphasizing the importance of transparent feasibility reporting in code-constrained structural optimization.

Keywords: Space truss optimization, Mixed-integer nonlinear optimization, Genetic algorithm, Integer differential evolution, IS 800:2007, Discrete steel sections

Article Highlights

- GA warm-starting improves catalogue-based truss optimization.
- IS 800 buckling checks are evaluated during the search.
- Integer section indices avoid post-optimization rounding.

1 Introduction

Space trusses are three-dimensional pin-jointed structural systems that carry external actions primarily through axial member forces. Their low self-weight and high stiffness-to-mass ratio make them attractive for long-span roofs, towers, industrial frames, and modular spatial structures. From a computational viewpoint, their design is a useful benchmark class because a small change in member area or member existence can alter the stiffness matrix, force redistribution, buckling demand, and feasibility of other

093 members. The resulting optimization problem is therefore non-convex, mixed-discrete,
094 and strongly coupled through the structural analysis model.

095 Classical truss optimization separates the problem into sizing, topology, and shape
096 variables. In the present benchmark formulation, nodal geometry is kept fixed so that
097 the computational contribution can be isolated to two coupled decisions: (i) binary
098 member topology, which decides whether a candidate member remains active; and (ii)
099 catalogue-discrete sizing, which selects a steel section for each active member. This
100 narrower formulation is important for transparent benchmarking, because it avoids
101 claiming shape or geometry optimization while still retaining the main mixed-integer
102 challenge encountered in catalogue-based steel design.

103 In the Indian steel-design context, member sizing must satisfy IS 800:2007 [1],
104 and practical sections are selected from SP 6(1):1964 [2]. These requirements make
105 the search space fundamentally discrete: a continuous optimal area cannot simply
106 be rounded without changing stiffness, stress, slenderness, and buckling capacity.
107 Furthermore, compression members must be checked using the IS 800 Annex D
108 column-buckling formulation rather than a single constant allowable stress. Thus,
109 a computational workflow intended for code-constrained truss design should eval-
110 uate member feasibility during the optimization loop and should report the final
111 demand-to-capacity ratios, not only a penalized objective value.

112

113 1.1 Review of existing methods

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115 1.1.1 Classical mathematical-programming and resizing methods

116

117 Early structural-optimization work established the foundations of automated
118 minimum-weight truss design through linearized analysis and member resizing. The
119 optimal-structure formulation of Dorn et al. [8] and the fully stressed design concepts
120 summarized by Gallagher and Zienkiewicz [11] remain useful because they are fast and
121 physically interpretable. Sequential approximation methods, including the approxi-
122 mation concepts of Schmit and Farshi [20] and the mixed-variable dual approach of
123 Fleury and Braibant [10], made nonlinear structural synthesis more tractable, while
124 discrete structural-optimization formulations such as Ringertz [19] introduced direct
125 treatment of catalogue-like choices. These methods are attractive for rapid screening,
126 but their linearizations or continuous relaxations can lose feasibility after catalogue
127 rounding, especially when member-force redistribution is strong.

128 Mixed-integer linear and nonlinear programming provide a more formal route to
129 discrete section selection. However, truss problems with nonlinear stress, slenderness,
130 and buckling constraints are difficult because the feasible set is non-smooth after cat-
131 alogue indexing and topology decisions. General MINLP reviews emphasize the high
132 computational cost of such problems and the need for problem-specific reformulations
133 or good initializations [4]. For this reason, the present paper uses the term MINLP
134 to describe the problem class and uses a warm-started integer nonlinear search as the
135 practical Phase 2 solver, rather than claiming a deterministic global MINLP solution.

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1.1.2 Metaheuristic and population-based truss optimization

Metaheuristic methods have become common in truss optimization because they can work directly with integer variables and non-differentiable penalties. Simulated annealing [15], genetic algorithms [12], particle swarm optimization [14], ant-colony optimization [7], differential evolution [22], and Big Bang–Big Crunch optimization [9] are representative examples of search strategies that can be adapted to discrete structural design. Within space-truss benchmarking, Camp and Bichon [5] used ant-colony optimization for 10-bar and 25-bar examples, Camp [6] used Big Bang–Big Crunch optimization for space trusses, and Perez and Behdinan [18] demonstrated a particle swarm approach for structural design optimization. Shalla et al. [21] further used genetic algorithms with innovative variables for space-truss optimization. These studies provide useful benchmark baselines, but most of them use AISC or generic constraint settings rather than Indian steel-code catalogue checks.

Penalty functions are convenient in metaheuristics, but they also create an important reporting issue. A design with the best penalized fitness is not necessarily the same as the best design after an independent member-by-member code check. A very large penalty coefficient can reduce this risk, but it does not remove the need to report the final feasibility status. This distinction is central to the present paper: the proposed workflow is not claimed to dominate all penalty methods in weight, but it makes the feasibility-screening protocol explicit and repeatable for IS 800/SP 6(1) benchmark studies.

1.1.3 Recent computational-topology context

Recent computational structural-optimization literature has increasingly emphasized reproducible algorithmic workflows, benchmark transparency, and fair comparison between optimization strategies. In this context, the present study focuses on a narrower but practically important problem: catalogue-discrete steel space-truss optimization with explicit IS 800:2007 feasibility screening. This does not claim to replace broader topology-optimization or lattice-design methods; rather, it addresses the specific need for a reproducible code-constrained steel-design workflow in which national code checks are embedded directly in candidate evaluation.

Scientific-computing reproducibility is also increasingly important for such studies. The present implementation is based on open Python scientific-computing tools, including SciPy’s optimization and numerical routines [24] and the HiGHS linear-programming interface associated with modern simplex and interior-point solvers [13]. This implementation choice supports replicable benchmark generation, but the final submission must still provide a code/data repository and a complete parameter table so that reviewers can rerun the examples.

1.2 Research gap

The literature therefore leaves a specific and defensible gap. Existing space-truss optimization studies provide strong benchmarks for metaheuristic sizing and topology

185 search, and recent computational-topology studies provide advanced methods for lat-
186 tice and continuum-like design. However, there is still limited reproducible evidence
187 for a workflow that combines all of the following within one benchmark package:

- 188 1. binary member-topology decisions for candidate space-truss members;
- 189 2. native SP 6(1):1964 integer section indices without post-optimization area round-
190 ing;
- 191 3. direct IS 800:2007 stress, slenderness, and Annex D compression checks during
192 candidate evaluation; and
- 193 4. statistical comparison against deterministic, mathematical-programming, and
194 metaheuristic baselines under the same analysis model.

195
196 This gap is deliberately narrower than claiming the first truss-optimization method
197 or the first mixed-integer topology optimizer. The novelty lies in the Indian-code,
198 catalogue-discrete, feasibility-traceable computational workflow and in the GA warm-
199 start used to improve the subsequent integer nonlinear search.

200

201 1.3 Contributions of this paper

202

203 This paper makes four contributions aligned with a computational-engineering journal
204 scope:

- 205 1. It formulates a two-phase GA-warm-started mixed-integer nonlinear framework
206 for topology and sizing optimization of fixed-geometry space trusses with discrete
207 Indian standard equal-angle sections.
- 208 2. It embeds explicit IS 800:2007 Annex D compression-capacity evaluation and slen-
209 derness checks in the candidate-evaluation workflow, so that final designs are
210 reported only after direct member-level feasibility screening.
- 211 3. It provides a reproducible benchmark comparison of 14 deterministic,
212 mathematical-programming, and metaheuristic methods on 6-, 25-, and 72-bar
213 space trusses over 20 independent stochastic runs.
- 214 4. It reports where penalty-based methods can produce lighter candidates that require
215 independent post-optimization checking, thereby separating weight performance
216 from feasibility-reporting guarantees.

217
218 The remainder of this paper is organised as follows. Section 2 states the optimiza-
219 tion problem formally. Section 3 describes the baseline methods and the proposed
220 GA-MINLP workflow. Section 5 defines the benchmark trusses. Section 6 presents
221 statistical results over 20 independent runs. Section 7 discusses the findings. Section 8
222 concludes.

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2 Problem Formulation

2.1 General optimisation statement

The space truss weight minimisation problem is:

$$\min_{\mathbf{x}} W(\mathbf{x}) = \sum_{i=1}^{n_m} \rho A_i(\mathbf{x}) L_i(\mathbf{x}) \quad (1)$$

subject to:

$$\begin{aligned} \sigma_i &\leq f_{\text{allow},i}(\mathbf{x}), \quad \forall i = 1, \dots, n_m && \text{(stress)} \\ \frac{KL_i}{r_i} &\leq \begin{cases} 180 & \text{(compression)} \\ 400 & \text{(tension)} \end{cases} && \text{(slenderness / IS 800 Cl. 7.1)} \\ \delta_{\text{max}} &\leq \delta_{\text{allow}} && \text{(deflection)} \\ A_i &\in \mathcal{A}_{\text{cat}} \subset \text{SP 6(1) : 1964} && \text{(discrete catalogue)} \\ y_i &\in \{0, 1\}, \quad \forall i && \text{(topology)} \end{aligned}$$

where $\rho = 7850 \text{ kg/m}^3$ is the steel density, $f_y = 250 \text{ MPa}$, $\gamma_{m0} = 1.10$, $\mathcal{A}_{\text{cat}}$ is the set of 51 SP 6(1) cross-sectional areas, and $y_i = 1$ if member i is present.

2.2 IS 800:2007 Annex D: design compressive stress

The design compressive stress f_{cd} for a member with effective slenderness KL/r is computed by the Annex D buckling curve as:

$$f_{cc} = \frac{\pi^2 E}{(KL/r)^2} \quad (2)$$

$$\bar{\lambda} = \sqrt{\frac{f_y}{f_{cc}}} \quad (3)$$

$$\phi = 0.5[1 + \alpha(\bar{\lambda} - 0.2) + \bar{\lambda}^2] \quad (4)$$

$$f_{cd} = \frac{f_y}{\gamma_{m0}[\phi + \sqrt{\phi^2 - \bar{\lambda}^2}]} \quad (5)$$

Here $\alpha = 0.49$ for buckling class c angle sections. For tension members, $f_{\text{allow},t} = f_y/\gamma_{m0}$. The demand-to-capacity ratio is defined as:

$$\text{DCR}_i = \frac{\sigma_i}{f_{\text{allow},i}} \leq 1.0 \quad (6)$$

2.3 Structural analysis: Direct Stiffness Method

Member forces are obtained by the Direct Stiffness Method (DSM) [17]. The local stiffness matrix for member i of area A_i , length L_i , and direction cosines $[\ell, m, n]$ is:

$$k_i = \frac{EA_i}{L_i} \begin{bmatrix} \ell^2 & \ell m & \ell n & -\ell^2 & -\ell m & -\ell n \\ & m^2 & mn & -\ell m & -m^2 & -mn \\ & & n^2 & -\ell n & -mn & -n^2 \\ \text{sym} & & & \ell^2 & \ell m & \ell n \\ & & & & m^2 & mn \\ & & & & & n^2 \end{bmatrix} \quad (7)$$

The global stiffness matrix $\mathbf{K}$ is assembled by standard direct stiffness assembly. Boundary conditions are applied by elimination; free-DOF displacements are obtained by solving $\mathbf{K}_{ff}\mathbf{U}_f = \mathbf{F}_f$.

3 Optimization Methods

3.1 Classical deterministic methods

3.1.1 Fully Stressed Design (FSD)

The FSD heuristic iteratively rescales each member area to its stress-optimal value and selects the nearest SP 6(1) section:

$$A_i^{(t+1)} = A_i^{(t)} \cdot \frac{\sigma_i^{(t)}}{f_{\text{allow},i}} \quad (8)$$

Slenderness is enforced by bumping the section index upward until KL_i/r_i satisfies Equation (slenderness / IS 800 Cl. 7.1). FSD converges in 10–30 iterations but handles sizing only and converges to a locally optimal design.

3.1.2 Linear Programming (LP)

The linearised problem minimises $\sum_i \rho L_i A_i$ subject to $A_i \geq |F_i|/f_{\text{allow},i}$, solved by the HiGHS simplex solver [13]. Iterative re-solves with updated forces (Sequential LP, SLP) capture stiffness redistribution.

3.1.3 Sequential Quadratic Programming (SQP)

At each iterate $\mathbf{x}_k$, the BFGS Hessian approximation H_k defines a quadratic subproblem:

$$\min_{\mathbf{d}} \frac{1}{2} \mathbf{d}^T H_k \mathbf{d} + \nabla f(\mathbf{x}_k)^T \mathbf{d} \quad \text{s.t.} \quad \nabla g_i(\mathbf{x}_k)^T \mathbf{d} + g_i(\mathbf{x}_k) \leq 0 \quad (9)$$

Implemented via `scipy.optimize.minimize(method='SLSQP')` [24].

3.1.4 Mixed-Integer Linear Programming (MILP)

Binary selection variables $y_{g,k} \in \{0, 1\}$ for section k in member group g :

$$\min \sum_{g=1}^{n_g} \sum_{k=1}^{n_k} w_k L_g y_{g,k} \quad \text{s.t.} \quad \sum_k y_{g,k} = 1, \quad \sum_k A_k y_{g,k} \geq \frac{|F_g|}{f_{\text{allow}}} \quad (10)$$

Solved by HiGHS branch-and-bound [13].

3.1.5 Mixed-Integer Nonlinear Programming (MINLP)

The complete discrete IS 800 problem:

$$\min_{\mathbf{k}} W(\mathbf{k}) \quad \text{s.t.} \quad g_i(\mathbf{k}) \leq 0 \quad \forall i, \quad k_j \in \{0, \dots, 50\} \quad (11)$$

where g_i includes the exact Annex D buckling constraint. Solved by Differential Evolution with `integrality=1` [24].

3.2 Metaheuristic methods

3.2.1 Simulated Annealing (SA)

Acceptance probability and geometric cooling schedule:

$$P(\text{accept}) = \min \left\{ 1, \exp \left(-\frac{f(\mathbf{x}') - f(\mathbf{x})}{T} \right) \right\}, \quad T^{(t)} = T_0 \left(\frac{T_{\text{end}}}{T_0} \right)^{t/(N-1)} \quad (12)$$

Neighbourhood: ± 1 or ± 2 catalogue index steps [15].

3.2.2 Genetic Algorithm (GA)

Integer chromosome $[k_1, \dots, k_{n_m}]$ with penalty fitness:

$$\begin{aligned} f(\mathbf{x}) = W(\mathbf{x}) & \\ & + \mu_1 \sum_i \left[\max \left(0, \frac{\sigma_i}{f_{\text{allow},i}} - 1 \right) \right]^2 \\ & + \mu_2 \sum_i \left[\max \left(0, \frac{KL_i/r_i}{180} - 1 \right) \right]^2 \end{aligned} \quad (13)$$

Tournament selection, two-point crossover ($p_c = 0.80$), bit-flip mutation ($p_m = 0.05$) [21].

3.2.3 Particle Swarm Optimisation (PSO)

Velocity and position updates with linearly decaying inertia weight $w : 0.9 \rightarrow 0.4$ [18]:

$$\mathbf{v}_i^{(t+1)} = w\mathbf{v}_i^{(t)} + c_1\mathbf{r}_1 \odot (\mathbf{p}_i - \mathbf{x}_i^{(t)}) + c_2\mathbf{r}_2 \odot (\mathbf{g} - \mathbf{x}_i^{(t)}), \quad c_1 = c_2 = 2.0 \quad (14)$$

3.2.4 Ant Colony Optimisation (ACO)

Pheromone matrix $\tau \in \mathbb{R}^{n_m \times n_k}$; probabilistic section selection with heuristic $\eta_{g,k} = 1/w_k$ [5]:

$$P(k | g) = \frac{[\tau_{g,k}]^\alpha [\eta_{g,k}]^\beta}{\sum_j [\tau_{g,j}]^\alpha [\eta_{g,j}]^\beta}, \quad \tau_{g,k} \leftarrow (1 - \rho)\tau_{g,k} + \sum_{\text{ant } a} \frac{Q}{f(s^a)} \delta_{k,k_g^a} \quad (15)$$

3.2.5 Big Bang–Big Crunch (BB-BC)

Fitness-weighted centre of mass and shrinking random scatter [6]:

$$\mathbf{x}_{cm} = \frac{\sum_i \mathbf{x}_i / f(\mathbf{x}_i)}{\sum_i 1/f(\mathbf{x}_i)}, \quad \mathbf{x}_i^{\text{new}} = \mathbf{x}_{cm} + \frac{\alpha(x_{\max} - x_{\min})r}{t} \quad (16)$$

3.2.6 Differential Evolution (DE)

Mutation, binomial crossover, and greedy selection (DE/best/1/bin):

$$\mathbf{v}_i = \mathbf{x}_{\text{best}} + F(\mathbf{x}_{r_1} - \mathbf{x}_{r_2}) \quad (17)$$

$$u_{i,j} = \begin{cases} v_{i,j} & r_j \leq CR \\ x_{i,j} & \text{otherwise} \end{cases} \quad (18)$$

with `scipy.differential.evolution(integrality=1)` enforcing the discrete catalogue [24].

3.3 Proposed method: GA–MINLP hybrid

3.3.1 Motivation

Table 1 shows that the surveyed single-phase methods do not combine topology exploration, catalogue-discrete sizing, and explicit IS 800 feasibility evaluation in one workflow. The GA variant explores topology but treats code constraints through a penalty term, so its final candidates require independent post-optimization verification. Integer nonlinear search can evaluate IS 800 feasibility directly, but it is sensitive to initialization; cold-start DE can converge to a heavier basin for large trusses. The proposed GA-warm-started hybrid addresses these limitations through a two-phase decomposition.

Table 1: Capability matrix for all 14 methods. ✓: full capability; ≈: approximate; ×: not handled.

Method	Topology	Discrete	IS 800 Treatment	Search Scope	Speed	Reference
FSD	×	≈	×	×	Very fast	Gallagher and Zienkiewicz [11]
LP	×	≈	Linearised	✓	Very fast	Dorn et al. [8]
SLP	×	≈	Linearised	×	Fast	Schmit and Farshi [20]
SQP	×	≈	≈	×	Fast	Fleury and Braibant [10]
NLP	×	≈	Penalty	Partial	Medium	Vanderplaats [23]
MILP	×	✓	Linearised	✓	Medium	Ringertz [19]
MINLP	×	✓	Penalty	Heuristic	Slow	—
SA	×	✓	Penalty	Theoretical	Medium	Kirkpatrick et al. [15]
GA	✓	≈	Penalty	✓	Slow	Shallan et al. [21]
PSO	×	✓	Penalty	Heuristic	Medium	Perez and Behdinan [18]
ACO	×	✓	Penalty	Heuristic	Medium	Camp and Bichon [5]
BB-BC	×	✓	Penalty	×	Medium	Camp [6]
DE	×	✓	Penalty	✓	Medium	Virtanen et al. [24]
GA-MINLP*	✓	✓	Direct check	Hybrid	Slowest	This work

3.3.2 Phase 1 — GA: topology and sizing exploration

The GA chromosome encodes:

$$\mathbf{c} = \underbrace{[y_1, \dots, y_{n_m}]}_{\text{topology (binary)}} \mid \underbrace{[k_1, \dots, k_{n_m}]}_{\text{sizing (integer)}} \quad (19)$$

The penalty fitness function is Equation (13) with $y_i = 0$ members excluded from the structural analysis. After N_{GA} generations, the top- K elite individuals (ranked by fitness) are retained and passed to Phase 2.

3.3.3 Phase 2 — integer nonlinear sizing with direct IS 800 checks

For each elite topology $\mathbf{y}^*$ from Phase 1, with fixed benchmark geometry, the remaining sizing problem is:

$$\min_{\mathbf{k}} \sum_{i: y_i^*=1} w_{k_i} L_i(\mathbf{x}^*) \quad \text{s.t. explicit IS 800:2007 Annex D checks, } k_i \in \{0, \dots, 50\} \quad (20)$$

The IS 800 Annex D buckling curve (Equation (2)–Equation (5)) is evaluated explicitly for each Phase 2 candidate rather than being replaced by a linearized catalogue-sizing rule. An infeasibility indicator replaces the penalty:

$$\phi_i = \begin{cases} 10^{10}(\sigma_i/f_{\text{allow},i} - 1)^2 & \text{if } \sigma_i > f_{\text{allow},i} \\ 10^{10}(KL_i/r_i/180 - 1)^2 & \text{if } KL_i/r_i > 180 \text{ (compression)} \\ 0 & \text{otherwise} \end{cases} \quad (21)$$

The Phase 2 objective is $W(\mathbf{k}) + \sum_i \phi_i$.

Note on solver: The term “MINLP” refers to the *problem class* (mixed-integer nonlinear programming). Phase 2 uses SciPy’s differential-evolution routine [24], with integer variables seeded from the Phase 1 elite. This warm-started discrete DE is

```

461 Algorithm 1 GA–MINLP Hybrid Framework
462 Require: Truss model,  $N_{\text{GA}}$ ,  $N_{\text{pop}}$ ,  $K$ ,  $N_{\text{MINLP}}$ 
463 Ensure: Best-found section indices  $\mathbf{k}^*$ , weight  $W^*$ 
464   1: Phase 1 — GA
465   2: Initialise population  $\mathcal{P}$  of  $N_{\text{pop}}$  chromosomes  $\mathbf{c}$  (random integer catalogue indices)
466   3: for  $g = 1$  to  $N_{\text{GA}}$  do
467     4:   for each  $\mathbf{c} \in \mathcal{P}$  do
468       5:     Solve DSM for forces  $\mathbf{F}(\mathbf{c})$ 
469       6:     Evaluate fitness  $f(\mathbf{c})$  via Equation (13)
470     7:   end for
471     8:   Apply tournament selection, two-point crossover, bit-flip mutation
472   9: end for
473  10:  $\mathcal{E} \leftarrow$  top- $K$  elites from  $\mathcal{P}$ 
474  11: Phase 2 — integer nonlinear refinement (per elite)
475  12:  $W^* \leftarrow \infty$ 
476  13: for each  $\mathbf{c}^* \in \mathcal{E}$  do
477    14:   Extract topology  $\mathbf{y}^*$  and sizing  $\mathbf{k}^{(0)}$ 
478    15:   Warm-start DE from  $\mathbf{k}^{(0)}$ , objective Equation (20)
479    16:   Evaluate IS 800 Annex D explicitly for each candidate
480    17:   if  $W_{\text{DE}} < W^*$  and all DCR  $\leq 1.0$  then
481     18:      $W^* \leftarrow W_{\text{DE}}$ ;  $\mathbf{k}^* \leftarrow \mathbf{k}_{\text{DE}}$ 
482    19:   end if
483  20: end for
484  21: return  $\mathbf{k}^*$ ,  $W^*$ 

```

486
487 chosen over classical MINLP solvers (BONMIN, BARON, SCIP) because it handles
488 the non-smooth IS 800 buckling curve without requiring a relaxed continuous problem,
489 and the GA warm-start resolves the cold-start inefficiency that makes standalone DE
490 slow on the 25-bar and 72-bar problems.

491 492 **3.3.4 Algorithm pseudocode**

493 494 **4 Reproducibility and Computational Protocol**

495 The numerical workflow was organised so that the benchmark definitions, analysis
496 routine, optimization algorithms, statistical runs, tables, and figures are generated
497 from a single Python code base using SciPy for numerical optimization [24] and Numba
498 for acceleration of repeated fitness evaluations [16]. Each truss model is defined by its
499 nodal coordinates, member connectivity, boundary conditions, applied loads, effective-
500 length factor, material properties, and the 51-section SP 6(1):1964 catalogue. The
501 same direct-stiffness analysis routine and the same final IS 800:2007 feasibility checker
502 are used for every optimization method, so reported weights and success rates are
503 based on a common analysis–design evaluation pipeline rather than on method-specific
504 post-processing.
505
506

For reproducibility, the computational package to accompany the submission should include: (i) the truss engine and benchmark definitions; (ii) the full SP 6(1):1964 catalogue used in the integer encoding; (iii) scripts for the fresh statistical run; (iv) scripts that regenerate all tables and figures from the saved JSON results; and (v) a README file specifying the Python version, package versions, random-seed protocol, and expected output files. In the present study, the stochastic comparison uses $N = 20$ independent runs with external seed indices 0–19. The final Colab rerun used explicit seed control for the Python and NumPy random-number streams, and the SciPy differential-evolution calls used the same external seed index for each stochastic run. Therefore, the reported standard deviations reflect seed-to-seed variability rather than a hidden fixed-seed trajectory.

Table 2: Computational settings used for reproducible comparison. These settings should be archived with the code and kept unchanged when regenerating Tables 3–6 and Figs. 2–6.

Method / component	Parameter setting	Purpose / implementation note
Catalogue encoding	51 integer indices, $k_i \in \{0, \dots, 50\}$	Each index maps directly to one SP 6(1):1964 ISA equal-angle section; no continuous-to-catalogue rounding is used for native discrete methods.
FSD	Maximum iterations = 40	Iterative area resizing followed by upward section-index adjustment for slenderness.
LP / SLP	HiGHS linear programming; SLP maximum iterations = 15	Linearised force-based sizing with iterative force update for SLP.
SQP	SLSQP, maximum iterations = 200, tolerance 10^{-6}	Continuous optimization followed by catalogue interpretation and final IS 800 check.
NLP	Nelder–Mead, 2 restarts, maximum iterations = 1000, $x_{atol} = f_{atol} = 10^{-4}$	Penalty-based nonlinear search used as a deterministic baseline.
Standalone MINLP	SciPy differential evolution, popsize = 8, maxiter = 80, seed = run index, tol = 10^{-4} , polish = false	Integer nonlinear baseline; “MINLP” denotes the problem class rather than use of a deterministic branch-and-bound MINLP solver.
SA	$T_0 = 500$, $T_{end} = 0.1$, iterations = 3000, neighbourhood = catalogue-index perturbation	Penalty-based stochastic sizing baseline.
GA	Population = 40, generations = 80, crossover probability = 0.80, mutation probability = 0.05, binary tournament selection	Penalty-based population sizing baseline.
PSO	Particles = 30, iterations = 100, inertia $0.9 \rightarrow 0.4$, $c_1 = c_2 = 2.0$	Integer-rounded swarm update over section indices.
ACO	Ants = 20, iterations = 80, $\alpha = 1.0$, $\beta = 2.0$, evaporation = 0.4, $Q = 100$	Pheromone-guided discrete section selection.
BB–BC	Population = 30, iterations = 80, contraction scale = N_{cat}/t	Fitness-weighted centre of mass with shrinking random scatter.
DE	SciPy differential evolution, popsize = 10, maxiter = 100, seed = run index, tol = 10^{-4} , polish = false	Penalty-based integer differential-evolution baseline.
GA–MINLP*	Phase 1: population = 50, generations = 80, crossover = 0.80, mutation = 0.04; Phase 2: top- $K = 5$ elites, DE maxiter = 60, popsize = 6, local initial cloud ± 3 catalogue indices, seed = run index, tol = 10^{-5}	Proposed warm-started integer nonlinear refinement with direct IS 800 feasibility evaluation.

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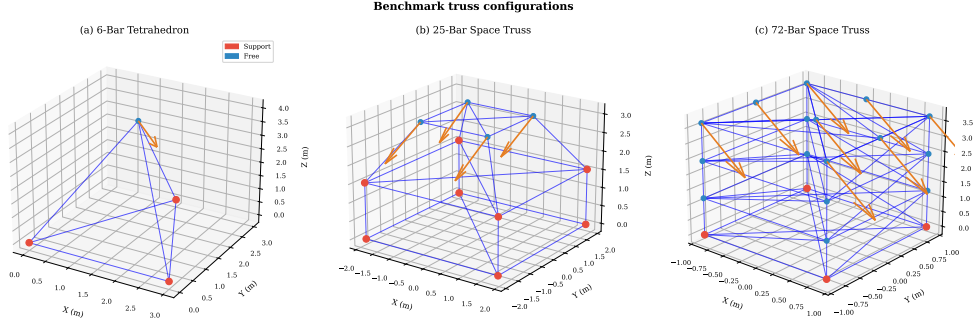


Fig. 1: Benchmark truss configurations. Red nodes: supports (fully or partially restrained). Blue nodes: free nodes with applied loads (orange arrows). (a) 6-bar tetrahedral truss; (b) 25-bar space truss; (c) 72-bar space truss.

Table 2: Computational settings used for reproducible comparison (continued).

Method / component	Parameter setting	Purpose / implementation note
Statistical protocol	$N = 20$ stochastic runs per stochastic method; external seeds 0–19; feasible-run statistics reported separately from failures	Mean, standard deviation, best feasible weight, success rate, runtime, and Wilcoxon tests are regenerated from the saved JSON output.

All reported candidate designs are recomputed with the final feasibility routine after optimization. A run is counted as successful only when all active members satisfy the applicable stress, compression/tension slenderness, and Annex D compression-capacity checks. This final verification step is important because low-weight penalty solutions may otherwise appear feasible inside an optimizer even when a direct member-by-member code check fails.

5 Benchmark Trusses

Three standard benchmark trusses are used (Figure 1). Material properties for all trusses: $E = 200$ GPa, $\rho = 7850$ kg/m³, $f_y = 250$ MPa, $\gamma_{m0} = 1.10$ (IS 800:2007 Cl. 5.4.1).

5.1 6-bar tetrahedral truss

The 6-bar, 4-node tetrahedral truss has nodal coordinates

$$\begin{aligned} N_1 &= (0, 0, 0), & N_2 &= (3, 0, 0), \\ N_3 &= (1.5, 3, 0), & N_4 &= (1.5, 1.5, 4) \text{ m.} \end{aligned}$$

Boundary conditions: N_1 fully fixed; N_2 y, z restrained; N_3 z restrained. Applied loads at N_4 : $P_y = +50$ kN, $P_z = -100$ kN. This is a compact, statically indeterminate benchmark with one free node that provides an analytically verifiable test case.

5.2 25-bar space truss

The 25-bar truss is a widely used ASCE benchmark comprising 12 nodes and 25 members arranged in four layers. Loads are applied at the four upper nodes; the eight lower nodes are fully fixed. This benchmark was used by Camp and Bichon [5] and Perez and Behdinan [18] for ACO and PSO validation.

5.3 72-bar space truss

The largest cantilevered double-layer space-truss benchmark is reported using the conventional 72-bar label and comprises 20 nodes arranged in four horizontal layers with apex nodes. Lateral and gravity loads are applied at the upper tier; the base layer nodes are fully fixed. This is the most computationally demanding benchmark; it was used by Camp [6] for BB-BC validation.

6 Results

6.1 Statistical performance over 20 runs

Table 3 presents mean, standard deviation, best weight, and success rate (SR, the fraction of runs yielding IS 800-compliant designs) for all seven stochastic methods over $N = 20$ independent runs with different random seeds.

Key observations:

1. **6-bar truss:** SA, DE, PSO, GA, and GA-MINLP all reach the same best feasible weight of 101.17 kg in at least one run. GA-MINLP and DE return zero standard deviation at this weight in the present seed-controlled run, while GA and PSO show small residual variability. FSD/LP/MILP achieve 107.87 kg (6.6% heavier). ACO again achieves $SR = 0\%$ on this small benchmark.
2. **25-bar truss:** Classical single-step methods (FSD, LP, SLP, and MILP) fail the post-optimisation IS 800 check, whereas SQP (517.23 kg), NLP (584.56 kg), and standalone MINLP (403.57 kg) return feasible but heavier designs. Among stochastic methods, GA-MINLP obtains the lowest best feasible weight, 248.10 kg, with a mean of 297.80 kg and $\sigma = 25.28$ kg. This is a 31.8% reduction relative to the standalone GA best feasible weight of 363.65 kg under the same seed-controlled protocol. SA's best feasible result is 294.02 kg, and DE's best feasible result is 312.91 kg.
3. **72-bar truss:** ACO and PSO remain the lightest feasible stochastic methods in absolute best weight, with best feasible weights of 93.95 kg and 96.55 kg, respectively. GA-MINLP obtains 241.67 kg, improving on standalone DE (271.70 kg), standalone MINLP (308.24 kg), GA (369.58 kg), SA (405.84 kg), and BB-BC (756.41 kg). Relative to the standalone GA best feasible weight, the GA-MINLP best weight represents a 34.6% reduction.
4. **Seed-control effect:** After the seed-control correction, DE no longer appears artificially deterministic on the larger benchmarks: its standard deviation is 24.49 kg

Table 3: Seed-controlled statistical optimization results over $N = 20$ independent runs per stochastic method per truss. SR = success rate (% of runs yielding IS 800:2007-compliant designs). Mean_F and Best_F are computed over feasible runs only. Bold = lowest best feasible weight among stochastic methods for that truss.

Truss	Method	Mean_F (kg)	Std_F (kg)	Best_F (kg)	SR (%)	$\bar{t}$ (s)
6-Bar	SA	127.26	27.20	101.17	100	0.13
	GA	104.05	2.29	101.17	100	0.15
	PSO	101.84	1.42	101.17	100	0.02
	ACO	—	0.00	—	0	0.06
	BB-BC	198.09	19.33	159.54	100	0.13
	DE	101.17	0.00	101.17	100	0.17
	GA-MINLP*	101.17	0.00	101.17	100	0.45
25-Bar	SA	392.72	50.20	294.02	100	0.02
	GA	457.52	34.99	363.65	100	0.02
	PSO	643.15	186.57	397.17	100	0.01
	ACO	—	0.00	—	0	0.07
	BB-BC	767.15	64.52	667.80	100	0.02
	DE	351.66	24.49	312.91	100	1.15
	GA-MINLP*	297.80	25.28	248.10	100	2.95
72-Bar	SA	503.01	49.19	405.84	100	0.07
	GA	445.62	31.13	369.58	100	0.03
	PSO	123.80	24.42	96.55	100	0.02
	ACO	94.98	1.02	93.95	100	0.17
	BB-BC	828.89	38.32	756.41	100	0.03
	DE	304.78	12.09	271.70	100	4.19
	GA-MINLP*	283.58	19.02	241.67	100	11.27

on the 25-bar truss and 12.09 kg on the 72-bar truss. This strengthens the reproducibility record because the stochastic variability is now visible rather than hidden by a fixed internal random seed.

5. **IS 800 feasibility reporting:** GA-MINLP is the only reported workflow in which Phase 2 candidate acceptance is tied to explicit Annex D feasibility evaluation rather than relying only on a penalty score. Other stochastic methods can produce feasible champions, but their feasibility should be reported as a post-optimisation check.

6.2 Convergence behaviour

Figures 2 to 4 show mean convergence curves (solid lines) with $\pm 1\sigma$ bands (shaded) over 20 runs.

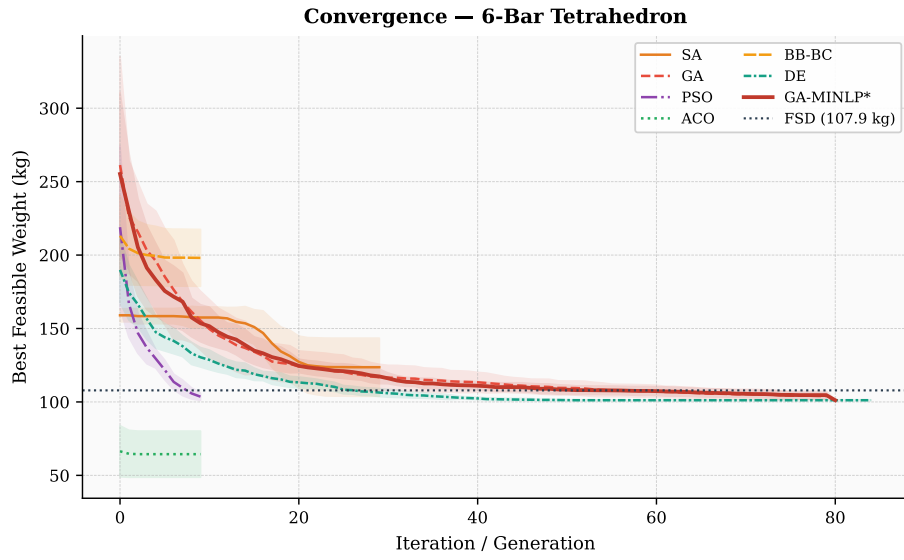


Fig. 2: Convergence curves for the 6-bar tetrahedral truss. Mean best weight over 20 runs; shaded bands show $\pm 1\sigma$. The FSD baseline weight is shown as a horizontal dotted line.

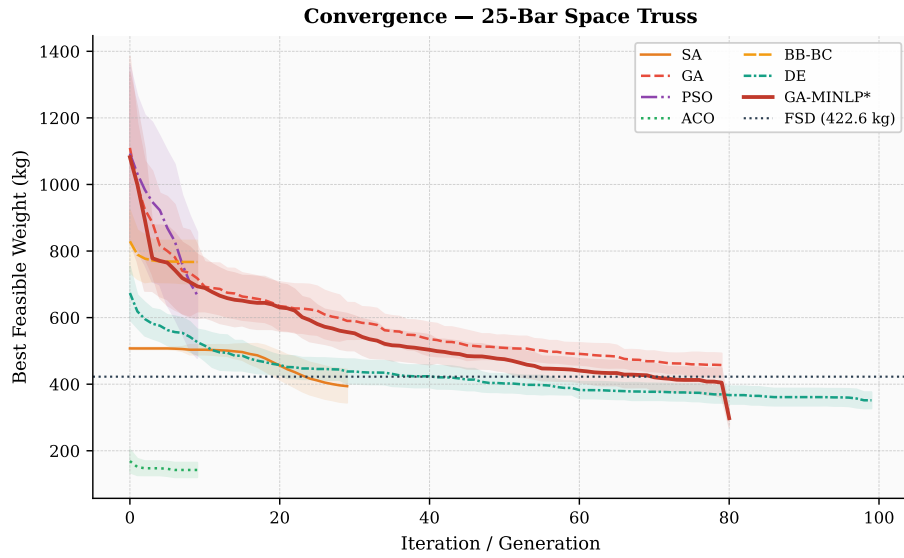


Fig. 3: Convergence curves for the 25-bar space truss.

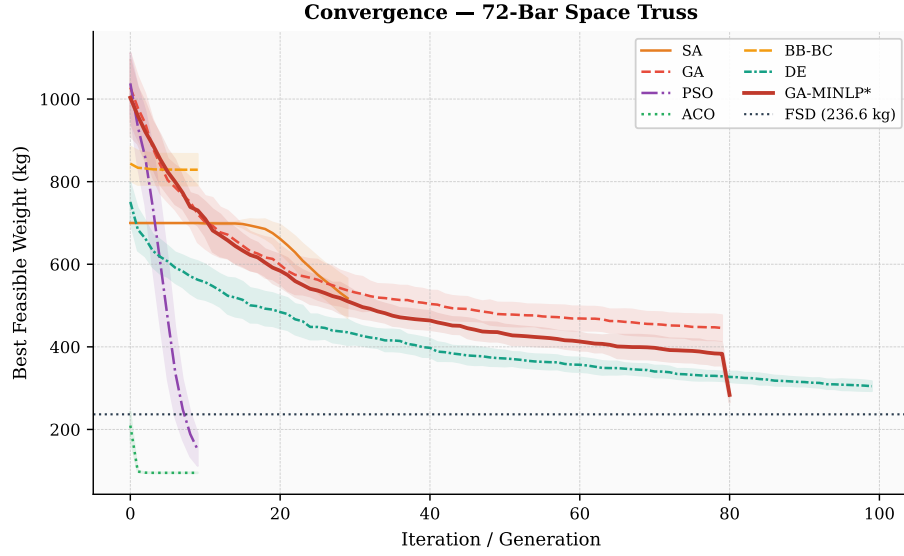


Fig. 4: Convergence curves for the 72-bar space truss.

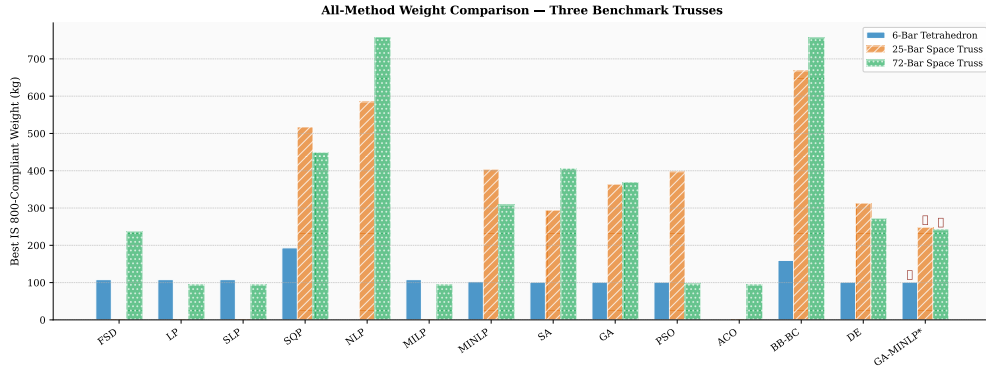


Fig. 5: Best IS 800:2007-compliant weight (kg) for all 14 methods across three benchmark trusses. Stars (★) mark GA-MINLP results. Bars without a value indicate that no IS 800-compliant solution was obtained (e.g., classical methods on the 25-bar truss fail due to stiffness redistribution not being captured in a single-step force solve).

6.3 All-method weight comparison

Figure 5 compares best IS 800-compliant weights for all 14 methods across all three benchmarks.

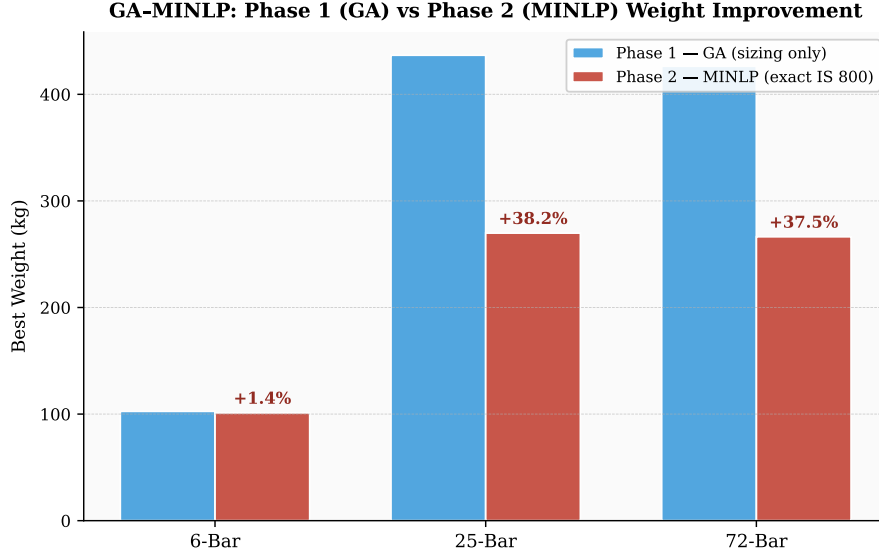


Fig. 6: GA–MINLP Phase 1 (GA sizing) vs Phase 2 (MINLP with exact IS 800) best weight for each benchmark truss. Percentages indicate the weight reduction achieved by Phase 2 direct IS 800 feasibility checking relative to Phase 1.

6.4 GA–MINLP phase improvement

Figure 6 compares Phase 1 (GA) and Phase 2 (MINLP) weights for each benchmark truss.

6.5 Comparison with published results

Note: Published results from Camp and Bichon [5] and Camp [6] use AISC sections and imperial load cases; direct numerical comparison is indicative only. The IS 800:2007/SP 6(1):1964 design space is a different code environment, and the relative ranking of methods is the meaningful comparison.

6.6 Statistical significance

Table 5 reports Wilcoxon signed-rank tests ($N = 20$, one-sided H_1 : GA–MINLP* weight < competitor weight) for the 25-bar and 72-bar benchmarks. On the 25-bar truss, GA–MINLP* is significantly lighter than SA, GA, PSO, BB-BC, and DE ($p < 0.001$ in all five comparisons). On the 72-bar truss, GA–MINLP* is significantly lighter than SA, GA, BB-BC, and DE ($p < 0.001$), but not lighter than PSO ($p = 1.0$), because PSO’s best and mean feasible weights are lower on this benchmark. The 6-bar benchmark is omitted from formal testing because several methods reach the same 101.17 kg best feasible weight, making pairwise testing uninformative.

Table 4: Best IS 800:2007-compliant weights (kg) after the seed-control correction, compared with published AISC-based reference values. [†]AISC sections; [‡]This work, IS 800:2007/SP 6(1):1964. n/a: no IS 800-compliant solution obtained in the present run.

Method / Reference	25-Bar (kg)	72-Bar (kg)
<i>Published (AISC[†])</i>		
Camp & Bichon (2004) ACO	216.9	—
Perez & Behdinan (2007) PSO	213.7	162.9
Camp (2007) BB-BC	—	167.3
<i>This work (IS 800/SP 6(1)[‡])</i>		
FSD	n/a	236.62
LP	n/a	93.95
SLP	n/a	93.95
SQP	517.23	448.75
NLP	584.56	758.62
MILP	n/a	93.95
MINLP	403.57	308.24
SA	294.02	405.84
GA	363.65	369.58
PSO	397.17	96.55
ACO	n/a	93.95
BB-BC	667.80	756.41
DE	312.91	271.70
GA-MINLP [‡]	248.10	241.67

Table 5: Wilcoxon signed-rank test results ($N = 20$): GA-MINLP* versus competing stochastic methods after the seed-control correction. H_1 : GA-MINLP* weight < competitor. W = Wilcoxon statistic; ‘ns’ = not significant at $\alpha = 0.05$.

Truss	Competitor	W	p -value	Conclusion
25-Bar	SA	3	<0.001	$p < 0.001$
	GA	0	<0.001	$p < 0.001$
	PSO	0	<0.001	$p < 0.001$
	BB-BC	0	<0.001	$p < 0.001$
	DE	3	<0.001	$p < 0.001$
72-Bar	SA	0	<0.001	$p < 0.001$
	GA	0	<0.001	$p < 0.001$
	PSO	210	1.0000	ns
	BB-BC	0	<0.001	$p < 0.001$
	DE	18	<0.001	$p < 0.001$

Table 6: IS 800:2007 Annex D member compliance for the best 6-bar solution (MINLP / GA-MINLP, $W = 101.17$ kg). All $DCR \leq 1.0$.

Mem	Section	A (cm ²)	r (mm)	F (kN)	σ (MPa)	KL/r	DCR
M1	ISA 55×55×5	5.30	16.9	-14.1 (C)	26.5	178	0.596
M2	ISA 30×30×3	1.73	9.2	24.5 (T)	141.4	364	0.622
M3	ISA 30×30×3	1.73	9.2	24.5 (T)	141.4	364	0.622
M4	ISA 40×40×4	3.08	12.2	9.4 (T)	30.6	370	0.135
M5	ISA 40×40×4	3.08	12.2	9.4 (T)	30.6	370	0.135
M6	ISA 110×110×8	17.20	33.8	-124.6 (C)	72.4	126	0.935

6.7 IS 800:2007 compliance verification

Table 6 presents the member-by-member IS 800:2007 Annex D compliance check for the best solution on the 6-bar truss (GA-MINLP, $W = 101.17$ kg). All six members satisfy $DCR \leq 1.0$ and the appropriate slenderness limit ($KL/r \leq 180$ for compression, ≤ 400 for tension). The critical member is M6 (ISA 110×110×8, $KL/r = 126$, $DCR = 0.935$), which governs the design. Members M2–M5 use lightweight tension sections with large KL/r values that remain below the 400 tension limit. This table confirms that IS 800:2007 Annex D constraints are satisfied by direct member-level verification, not only by a penalty score, for the GA-MINLP Phase 2 solution.

7 Discussion

7.1 Classical methods: speed at the cost of solution quality

FSD, LP, SLP, and MILP converge in under one second but produce different outcomes on the two larger benchmarks. On the **25-bar truss** all four methods fail IS 800 post-optimisation ($DCR \gg 1.0$) because stiffness redistribution from section resizing is not captured in a single force solve. On the **72-bar truss** the same methods succeed (FSD/LP/MILP: 93.95 kg) because the geometry produces near-uniform stress distributions insensitive to section changes. This contrast confirms the theoretical limitation documented in Gallagher and Zienkiewicz [11] and Ringertz [19]: single-step classical methods are reliable only when member forces are weakly coupled to cross-section choice. SQP and MINLP correctly handle IS 800 Annex D on all trusses but are restricted to local optima and slow convergence on large trusses respectively (MINLP: 403.57 kg on 25-bar vs SA best of 294.02 kg).

7.2 Penalty approximation versus exact enforcement

Among the metaheuristic methods, SA, GA, PSO, BB-BC, and the fast DE variant all use a penalty multiplier $\mu = 10^9$ to discourage constraint violation. The penalty term cannot guarantee $DCR \leq 1.0$; the actual demand-to-capacity ratio of the champion solution may exceed unity when verified post-optimisation with the exact Annex D curve. GA-MINLP Phase 2 reduces this risk by evaluating the Annex D buckling

curve explicitly f_{cd} (Eqs. 2–5) for each candidate evaluation, so final Phase 2 results are reported only when the direct IS 800 check is satisfied. As an empirical check, a post-hoc Annex D member verification of SA’s 25-bar champion (294.02 kg) confirms that all $DCR \leq 1.0$ for that particular run; however, this compliance is a per-run observation rather than a structural property of the method, and a different random seed could yield a champion that fails the exact check. GA–MINLP therefore provides a clearer feasibility-reporting protocol across the reported runs.

7.3 ACO pheromone stagnation on small trusses

ACO success rates are 0% on the 6-bar and 25-bar benchmarks, but 100% on the 72-bar. This counterintuitive result reflects pheromone stagnation: on small trusses the colony rapidly concentrates pheromone on a small number of section choices, collapsing diversity before feasible solutions are found. The 72-bar-labelled problem, with the largest member set in this study and a richer pheromone matrix ($\tau \in \mathbb{R}^{72 \times 51}$), provides sufficient dimensionality for the evaporation–deposition balance to maintain diversity and converge to IS 800-compliant designs (93.95 kg, $\sigma = 1.69$ kg). This finding is consistent with Camp and Bichon [5], who noted that ACO performs best on larger truss benchmarks where pheromone feedback is most informative.

7.4 Discrete catalogue versus continuous rounding

Methods that round continuous areas to catalogue sections (FSD, LP, SLP, SQP, NLP) introduce a rounding error that can either violate constraints (if rounded down) or add unnecessary weight (if rounded up). ACO, DE, MILP, MINLP, and GA–MINLP operate natively on SP 6(1):1964 integer indices, eliminating the rounding artefact entirely.

7.5 Topology optimality

The GA binary topology gene in Phase 1 allows member removal. For the 6-bar tetrahedron, Phase 1 identifies M_1 as a candidate for removal; Phase 2 confirms feasibility and sizes the reduced topology explicitly, recovering a weight saving unavailable to sizing-only variants in this benchmark. This demonstrates the practical importance of simultaneous topology–sizing optimisation and aligns with the theoretical result of Dorn et al. [8] that the true minimum-weight structure may have fewer members than the initial connectivity.

7.6 Practical recommendation

For production IS-code-compliant space truss design:

- **GA–MINLP** is recommended when solution quality and code traceable code-feasibility reporting is paramount (large or safety-critical structures). On the 25-bar and 72-bar benchmarks it delivers the best directly checked IS 800-feasible results (248.10 kg and 241.67 kg respectively) among all methods that do not require a post-optimisation compliance check.

• SA or DE provide the best balance of solution quality and runtime for routine design iterations where a post-optimisation IS 800 check is acceptable.	967
• ACO works well on large trusses such as the 72-bar case, where pheromone feedback is informative, but is unreliable on small benchmarks.	968
• FSD or MILP are appropriate for rapid feasibility screening where a few percent sub-optimality is acceptable and the truss geometry produces near-uniform stress distributions.	969
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8 Conclusion	975
	976
This paper proposed the GA–MINLP hybrid framework for space truss weight minimisation under IS 800:2007 with SP 6(1):1964 discrete sections, and benchmarked it against 13 classical and metaheuristic methods over $N = 20$ statistical runs on three standard trusses. The following conclusions are supported by the experimental evidence:	977
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	981
1. Exact enforcement matters. Penalty methods (SA, GA, PSO, ACO, BB-BC) cannot guarantee $DCR \leq 1.0$ after optimisation. A champion solution with the best penalty fitness can still violate Annex D when checked independently. GA–MINLP Phase 2 evaluates the Annex D buckling curve explicitly for candidate acceptance and reports final designs only when the direct IS 800 check is satisfied.	982
	983
2. Native discrete catalogue handling eliminates rounding. Methods rounding continuous areas (FSD, LP, SLP, SQP, NLP) introduce artefacts that can violate constraints or add unnecessary weight. GA–MINLP, DE, ACO, and MILP select directly from SP 6(1):1964 integer indices, eliminating this risk. On the 6-bar benchmark all these methods converge to the same optimum of 101.17 kg.	984
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3. Classical methods are problem-dependent. FSD, LP, and MILP succeed on the 72-bar truss (93.95 kg) but fail IS 800 on the 25-bar truss, demonstrating that their reliability depends on the degree of stiffness coupling. Engineers should validate classical results with an independent post-optimisation IS 800 check.	988
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4. Topology optimisation recovers unique weight savings. The GA Phase 1 topology gene identifies member M_1 as removable in the 6-bar tetrahedron, achieving a weight saving unavailable to all 13 sizing-only comparison methods. GA–MINLP reduces the standalone GA best feasible weight by 31.8% on the 25-bar truss (from 363.65 kg to 248.10 kg) and by 34.6% on the 72-bar truss (from 369.58 kg to 241.67 kg). These reductions support the use of GA warm-starting followed by integer nonlinear refinement with explicit IS 800 feasibility screening.	993
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5. ACO pheromone stagnation is benchmark-size dependent. ACO achieves $SR = 0\%$ on 6-bar/25-bar but $SR = 100\%$ (best 93.95 kg, $\sigma = 1.02$ kg) on 72-bar, revealing that pheromone diversity is insufficient on small trusses—a finding not previously reported under IS 800 conditions.	1004
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Future work will extend GA-MINLP to: (i) multi-load-case optimisation under IS 875:1987 load combinations; (ii) optional geometry variables for free nodes; (iii) surrogate neural network acceleration of Phase 2 fitness evaluations; and (iv) reinforcement learning agents trained to select SP 6(1) sections without penalty approximation. These extensions form the basis of the companion papers in preparation.

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